Name: Mico Angelo C. Olivan
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- **Data Structures and algorithms**
    - Data Structures involves storing, organizing, and processing data using structured formats. Meanwhile, algorithms are a step-by-step procedure. It makes sure specific problems are solved correctly.

- **Linear Data Structures**
    - It stores data in a linear order where each data component is connected. Because they are in a single level, traversing through every data piece is much easier.

        a. *One-Dimension Array*
            - It's a group of elements contained in a single row. The elements within can only be accessed by providing a distinctive index for each of them.

- **The Big O Notation**
    - It depicts how the spatial requirements of an algorithm scale with the input size. It analyses the algorithm's efficiency in a simple manner by focusing on key terms.

- **Basic Searching Algorithms**
    - It is useful for locating data within a dataset and retrieving information from previous data structures like arrays. It provides accuracy while also minimizing the time and resources needed.

        a. *Linear Search*
            - It's about scanning each element in a dataset until it chooses a desired element. Its implementation is much simpler and can work properly on sorted and unsorted datasets.

b. Binary Search

- It splits one dataset into two halves, compares both the target and centered element, concentrates on the important half based on the comparison, and then repeats the same process until the algorithm finds its target. It is much more efficient but it can only work on sorted datasets.

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